

```
In [1]: import pandas as pd
```

```
In [14]: df = pd.read_excel(r'C:\Users\MGAUTAM\OneDrive - United Nations\Data Science course
```

```
In [16]: df
```

```
Out[16]:
```

	IncumbIndexNumber	Duty Station	Duty Station Country	FirstName	LastName	Gender	Em Nati
0	10306828	New York	United States of America	THOMAS	FLETCHER	Male	Ki
1	644387	East Jerusalem	Occupied Palestinian Territory	Saad Khalidoun	K.B.ABDELHAQ	Male	P
2	367748	Nairobi	Kenya	Afifa	ISMAIL	Female	
3	142081	Amman	Jordan	Yahya	KHASBAH	Male	
4	71001597	Yangon	Myanmar	Shwe Cin	MYINT	Female	M
...	...	...	...	...	...	...	
2225	434333	New York	United States of America	ANNE SOFIE GARDE	THOMLE	Female	D
2226	107447	New York	United States of America	HELI	UUSIKYLA	Female	
2227	79975	Geneva	Switzerland	FITSUM	ASSEFA	Female	E
2228	587538	Kyiv	Ukraine	SARAH	HILDING DER WEDUWEN	Female	:
2229	887125	Geneva	Switzerland	James	WEATHERILL	Male	
2230 rows × 14 columns							
							

```
In [17]: len(df)
```

Out[17]: 2230

```
In [19]: df.shape
```

Out[19]: (2230, 14)

```
In [20]: df.columns
```

Out[20]: Index(['IncumbIndexNumber', 'Duty Station', 'Duty Station Country', 'FirstName', 'LastName', 'Gender', 'Employee Nationality', 'Regional Diversity', 'Contract Type', 'Personal Level', 'Branch Division', 'Contract End Date', 'EOD UN Secretariat', 'Employee Group'], dtype='object')

```
In [21]: len(df.columns)
```

Out[21]: 14

```
In [22]: df.isnull()
```

Out[22]:

	IncumbIndexNumber	Duty Station	Duty Station Country	FirstName	LastName	Gender	Employee Nationality
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...
2225	False	False	False	False	False	False	False
2226	False	False	False	False	False	False	False
2227	False	False	False	False	False	False	False
2228	False	False	False	False	False	False	False
2229	False	False	False	False	False	False	False

2230 rows × 14 columns

```
In [23]: df.isnull().sum()
```

```
Out[23]: IncumbIndexNumber      0
Duty Station                  0
Duty Station Country          0
FirstName                     1
LastName                      0
Gender                        1
Employee Nationality          1
Regional Diversity            1
Contract Type                 0
Personal Level                0
Branch Division               7
Contract End Date             0
EOD UN Secretariat           2
Employee Group                0
dtype: int64
```

```
In [28]: df_blank = df[df['FirstName'].isna()]
```

```
In [29]: df_blank
```

```
Out[29]:
```

	IncumbIndexNumber	Duty Station	Duty Station Country	FirstName	LastName	Gender	Employee Nationality
630	71410931	Yangon	Myanmar	NaN	ELIZABETH	Female	Myanmar

```
In [30]: df_blankName = df[df['FirstName'].isna()]
```

```
In [31]: df_blankGender = df[df['Gender'].isna()]
df_blankEmployeeNationality = df[df['Employee Nationality'].isna()]
df_blankEmployeeRegionalDiversity = df[df['Regional Diversity'].isna()]
```

```
In [33]: print (df_blankGender)
```

```
IncumbIndexNumber Duty Station      Duty Station Country      FirstName \
2222      507957      New York      United States of America      TEODOR STEFAN

LastName Gender Employee Nationality Regional Diversity Contract Type \
2222  GHERMAN      NaN      NaN      NaN      Fixed Term

Personal Level      Branch Division Contract End Date \
2222      01      Financing and Partnerships Division      2025-09-29

EOD UN Secretariat Employee Group
2222      2019-10-08      Benefits
```

```
In [34]: df.loc[(df['IncumbIndexNumber'] == 507957) &
               ((df['Gender'].isna()) | (df['Gender'] == "")), 'Gender'] = "Male"
```

```
In [36]: df.isnull().sum() # see bleow missing gender value has been updated. Now, no null f
```

```
Out[36]: IncumbIndexNumber      0
Duty Station                  0
Duty Station Country          0
FirstName                    1
LastName                     0
Gender                       0
Employee Nationality         1
Regional Diversity           1
Contract Type                0
Personal Level               0
Branch Division              7
Contract End Date            0
EOD UN Secretariat           2
Employee Group               0
dtype: int64
```

```
In [38]: df.head(3) # show first three records
```

```
Out[38]:
```

	IncumbIndexNumber	Duty Station	Duty Station Country	FirstName	LastName	Gender	Employ National
0	10306828	New York	United States of America	THOMAS	FLETCHER	Male	Unit Kingdo
1	644387	East Jerusalem	Occupied Palestinian Territory	Saad Khaloun	K.B.ABDELHAQ	Male	State Palest
2	367748	Nairobi	Kenya	Afifa	ISMAIL	Female	Ker

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```
In [40]: df.tail() #display last five records
```

Out[40]:

	IncumbIndexNumber	Duty Station	Duty Station Country	FirstName	LastName	Gender	Employ National
2225	434333	New York	United States of America	ANNE SOFIE GARDE	THOMLE	Female	Denm.
2226	107447	New York	United States of America	HELI	UUSIKYLA	Female	Finla
2227	79975	Geneva	Switzerland	FITSUM	ASSEFA	Female	Ethio
2228	587538	Kyiv	Ukraine	SARAH	HILDING DER WEDUWEN	Female	Swec
2229	887125	Geneva	Switzerland	James	WEATHERILL	Male	U

In [42]: `df.head()` # display first five records

Out[42]:

	IncumbIndexNumber	Duty Station	Duty Station Country	FirstName	LastName	Gender	Employ National
0	10306828	New York	United States of America	THOMAS	FLETCHER	Male	Unit Kingdo
1	644387	East Jerusalem	Occupied Palestinian Territory	Saad Khaldoun	K.B.ABDELHAQ	Male	State Palesti
2	367748	Nairobi	Kenya	Afifa	ISMAIL	Female	Ker
3	142081	Amman	Jordan	Yahya	KHASBAH	Male	Jorc
4	71001597	Yangon	Myanmar	Shwe Cin	MYINT	Female	Myann

In [49]: `df = df.sort_values(by="FirstName", ascending=False, na_position='first')`

In [50]: `df`

Out[50]:

	IncumbIndexNumber	Duty Station	Duty Station Country	FirstName	LastName	Gender	E Na
630	71410931	Yangon	Myanmar	NaN	ELIZABETH	Female	
756	71435821	Maiduguri	Nigeria	Zungwem Mary	TWAR	Female	
754	695199	Bamako	Mali	Zoumana	COULIBALY	Male	
504	71006759	Ouagadougou	Burkina Faso	Zoulfaou	DANKOURMA	Female	
1168	485913	Beirut	Lebanon	Zouheir	Bader	Male	
...	...	...	...	...	...	...	
954	10145724	New York	United States of America	*	KHAING SHWE SIN WIN	Female	
2167	627597	Mykolaiv	Ukraine	*	USAMA KABROUN AMIR SALIH	Male	
2056	71000089	Sana'a	Yemen	*	ABD AL- RAHIM HANI A OBALI	Male	
2005	312912	Kyiv	Ukraine	*	HTET HTET OO	Female	
1570	10055528	Riyadh	Saudi Arabia	*	MOHAMMED SELIMAN KHATER ZAID	Male	

2230 rows × 14 columns



In [51]: df['Duty Station Country']

```

Out[51]: 630          Myanmar
          756          Nigeria
          754          Mali
          504      Burkina Faso
          1168         Lebanon
          ...
          954  United States of America
          2167          Ukraine
          2056          Yemen
          2005          Ukraine
          1570      Saudi Arabia
          Name: Duty Station Country, Length: 2230, dtype: object

```

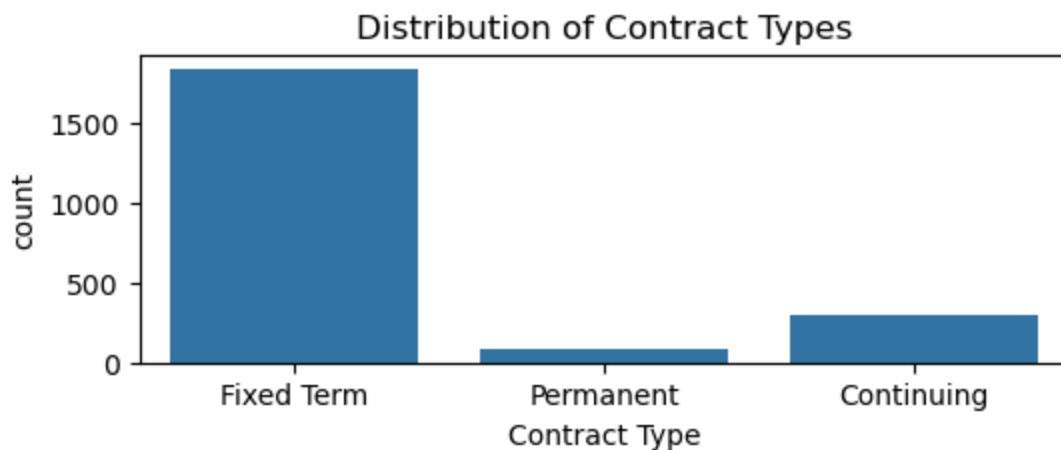
```
In [ ]:
```

```

In [60]: import matplotlib.pyplot as plt
import seaborn as sns

# plt.figure(figsize=(6,4)) # Adjust figure size
sns.countplot(x="Contract Type", data=df)
plt.title("Distribution of Contract Types")
plt.show()

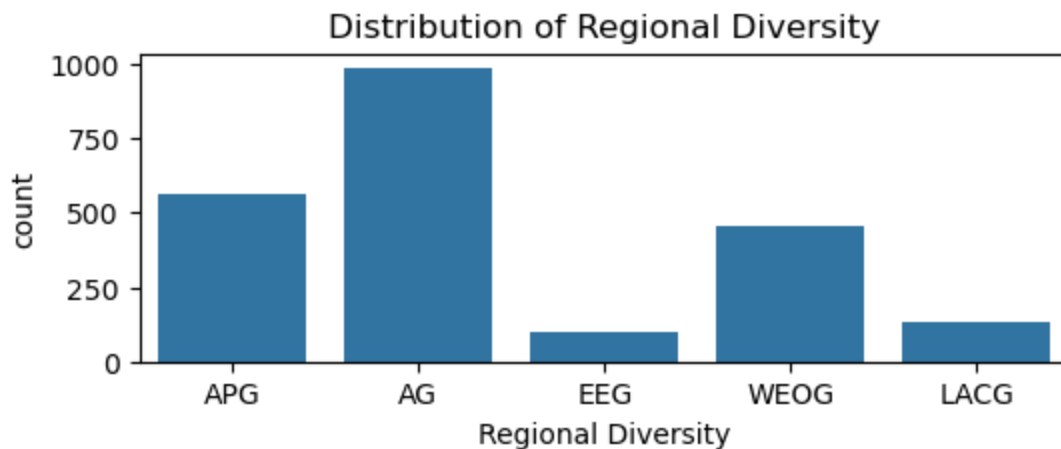
```



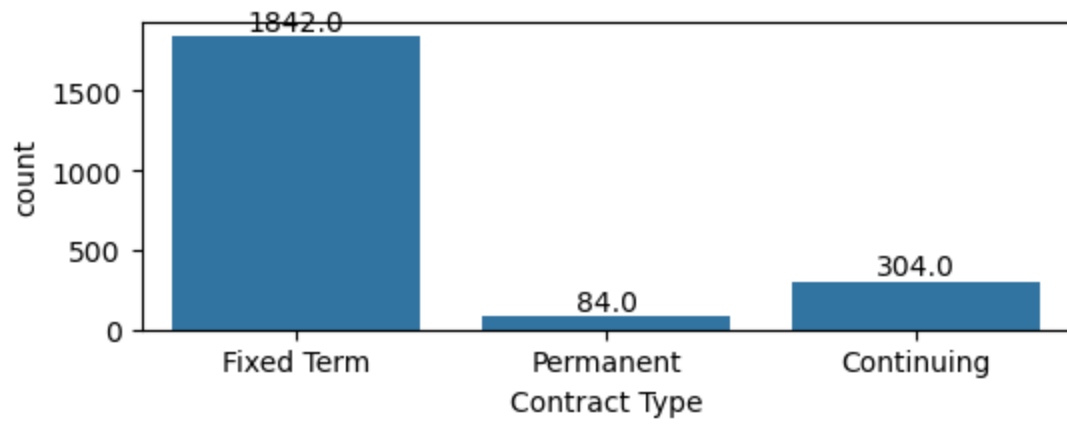
```

In [63]: sns.countplot(x="Regional Diversity", data=df)
plt.title("Distribution of Regional Diversity")
plt.show()

```



```
In [65]: ax = sns.countplot(x="Contract Type", data=df)
for p in ax.patches:
    ax.annotate(str(p.get_height()), (p.get_x() + p.get_width()/2., p.get_height()),
                ha='center', va='bottom')
plt.show()
```



In []: